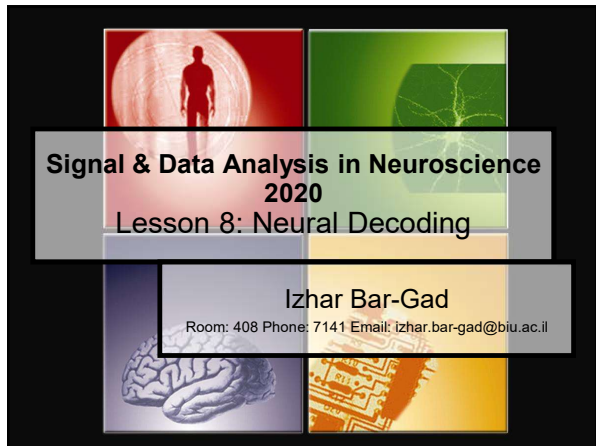


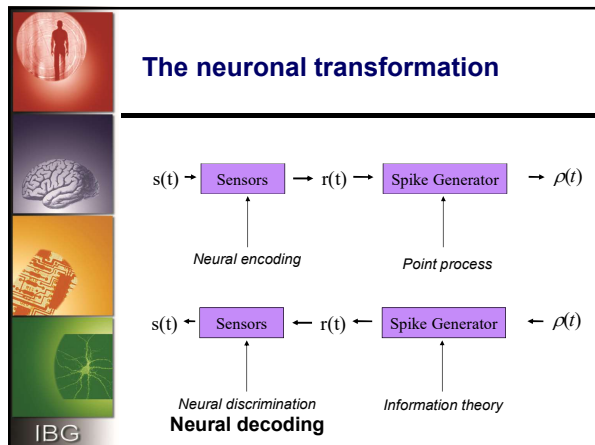


**Signal & Data Analysis in Neuroscience
2020**

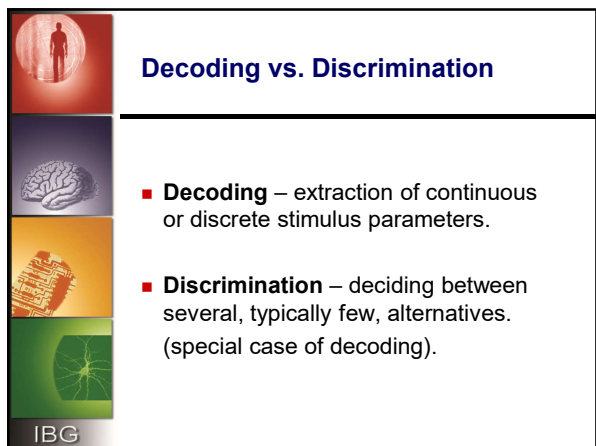
Lesson 8: Neural Decoding

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Decoding vs. Discrimination

- **Decoding** – extraction of continuous or discrete stimulus parameters.
- **Discrimination** – deciding between several, typically few, alternatives. (special case of decoding).

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Outline

- Population vectors
- Optimal estimators
- Estimation error

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Based upon: Theoretical neuroscience, *Dayan P & Abbott LF*, Ch. 3

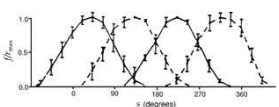
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Cricket cercal system I

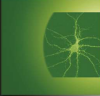
- Cricket cercal system
 - Hair cells generate spikes when deflected by wind
 - 4 inter-neurons receive input from thousands of hair cells connected to primary sensory neurons
- Maximal responses (**preferred directions**) of the neurons are spaced 90° apart.



From: Theunissen & Miller, 1991 and Salinas & Abbot, 1994

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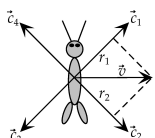


Cricket cercal system II

$$\left(\frac{f(s)}{r_{\max}}\right)_a = [\cos(s - s_a)]_+ = [\vec{v} \cdot \vec{c}_a]_+$$

s is the angle in degrees and $s_a, a = 1, \dots, 4$ is the preferred direction angle for each neuron. $\vec{v}$ is the wind direction and $\vec{c}_a$ is the preferred wind direction.

$$f_a(s) = \langle r_a(s) \rangle$$


Four interneurons are required instead of two because of the half-wave rectification and lack of representation of negative numbers.

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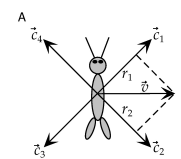
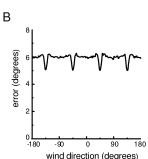



Cricket cercal system III

The animal can infer the wind direction from the inter-neuron responses by computing the population vector

$$\vec{v}_{pop} = \sum_{a=1}^4 \left(\frac{r}{r_{max}} \right)_a \vec{c}_a$$

The error $|\vec{s} - \vec{s}_{pop}|$ is about 6 degrees.

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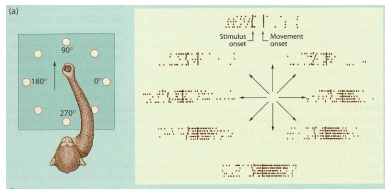
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Monkey primary motor cortex (M1) I

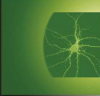
- The monkey is guided to move the lever from the center to one of eight peripheral locations during recording of neurons in M1.



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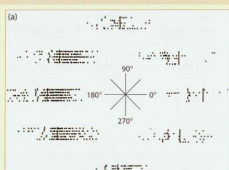
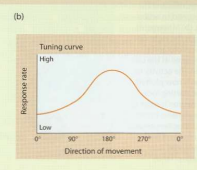
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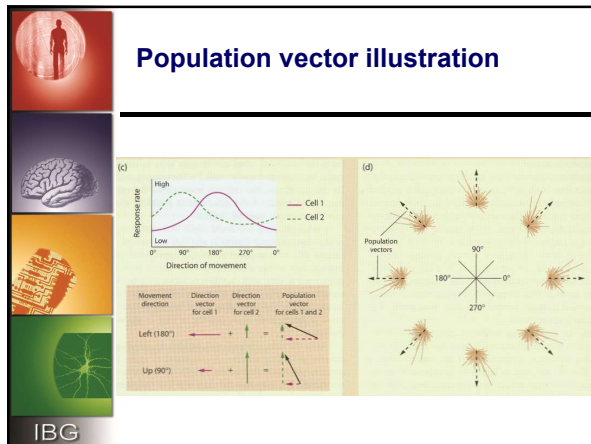
Preferred stimulus and tuning curve

- Preferred stimulus:** for each neuron, there is a stimulus value by which the neuron is mostly activated.
- Tuning curve:** the functional mapping between the neural activity (measured by mean firing rate) and the stimulus value.

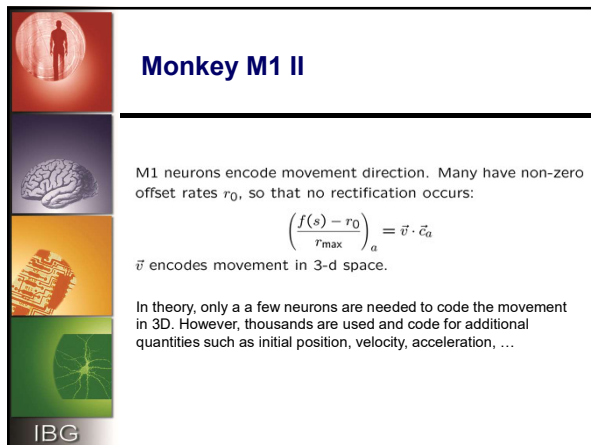



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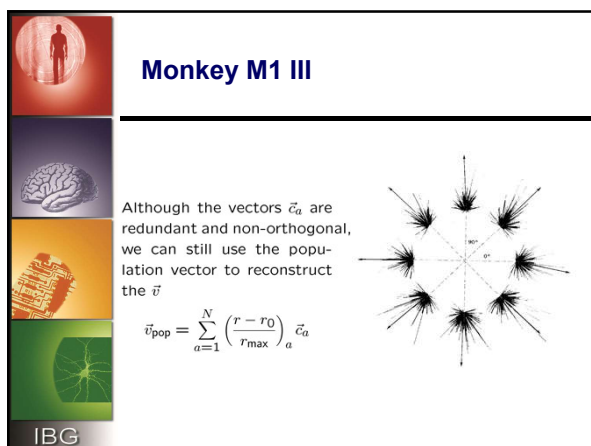
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Outline

- Population vectors
- Optimal estimators**
- Estimation error

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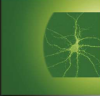
Optimal estimation of parameters

- Population vectors are easy to compute but are sub-optimal estimators in the general case.
- Given a probability model $p(s|r)$ and a given response vector r , we can compute estimators which follow **some** optimality rule:
 - MAP (maximum a posteriori)** estimator
 - ML (maximum likelihood)** estimator
 - Bayesian** estimator

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Optimization

- Optimization** - minimizing or maximizing a function by systematically choosing the values from within an allowed set.
- In our case we will try to minimize the **loss** (or cost) function which is a function of the sought decoding parameter/s.

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Likelihood - reminder

- A likelihood function is a conditional probability function considered a function of its **second** argument with its **first** argument held fixed. For a specific b the likelihood is $P(A|B=b)$.
- Probability allows us to predict unknown outcomes based on known parameters, while likelihood allows us to estimate unknown parameters based on known outcomes.
- It is also any other proportional function **$L(b|A) \propto P(A|B=b)$** for every $\alpha > 0$
- Thus, the numerical value of L is not important but only the **ratio** of likelihoods: $\frac{L(b_1|A)}{L(b_2|A)}$

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Bayes - reminder

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

x – cause y – effect

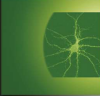
$$P(x|y) = \frac{P(y|x)P(x)}{\sum_x P(y|x)P(x)} \quad p(x|y) = \frac{p(y|x)p(x)}{\int_{-\infty}^{\infty} p(y|x)p(x)dx}$$

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}$$

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Bayesian decision making

- Suppose we know the prior distribution of possible values of $\theta \rightarrow p_0(\theta)$
- We also have a loss function $\lambda(\hat{\theta}, \theta)$ which measures the penalty for estimating $\hat{\theta}$ when actual value is θ
- The estimation problem is a Bayesian decision: choose the value of $\hat{\theta}$ which minimizes the risk given a set of n observations

$$R[\hat{\theta}|X^{(n)}] = \int p(\theta|X^{(n)})\lambda(\hat{\theta}, \theta)d\theta$$

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Maximum A-Posteriori (MAP)

- Let us look at $\lambda(\hat{\theta}, \theta) = \begin{cases} 0 & \text{if } \hat{\theta} = \theta \\ 1 & \text{if } \hat{\theta} \neq \theta \end{cases}$
 → the optimal estimator is the most likely value of θ given the data and $p_0(\theta)$
- In this case we care only about whether or not we made an error and not its **magnitude**.

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MAP estimation I

- The error function R is minimal for the most probable value of θ given the observation set X .
- This most likely value of the estimator:

$$\hat{\theta} = \arg \max_{\theta} p(\theta | X^{(n)}) = \arg \max_{\theta} \frac{p_0(\theta)p(X^{(n)} | \theta)}{p(X^{(n)})}$$

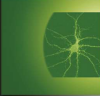
$$= \arg \max_{\theta} \frac{p_0(\theta)p(X^{(n)} | \theta)}{\int p(X^{(n)} | \theta')p_0(\theta')d\theta'}$$

argmax → the value of the given argument for which the value of the given expression attains its maximum value.

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MAP estimation II

- We can disregard the normalizing factor when looking for the maximum

$$\hat{\theta} = \operatorname{argmax}_{\theta} p_0(\theta)p(X^{(n)} | \theta)$$

when the data is i.i.d $p(X^{(n)} | \theta) = \prod_{i=1}^n p(x_i | \theta)$

$$\hat{\theta} = \operatorname{argmax}_{\theta} p_0(\theta) \prod_{i=1}^n p(x_i | \theta)$$

i.i.d. → independent and identically-distributed random variables.

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MAP estimation III

- We can assess the logarithm of the term as it is **monotonically increasing** with it.

$$\begin{aligned}\hat{\theta} &= \arg \max_{\theta} \left[p_0(\theta) \prod_{i=1}^n p(x_i | \theta) \right] \\ &= \arg \max_{\theta} \left(\log \left[p_0(\theta) \prod_{i=1}^n p(x_i | \theta) \right] \right) \\ &= \arg \max_{\theta} \left(\log p_0(\theta) + \log \prod_{i=1}^n p(x_i | \theta) \right) \\ &= \arg \max_{\theta} \left(\log p_0(\theta) + \sum_{i=1}^n \log p(x_i | \theta) \right)\end{aligned}$$

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MAP - the importance of priors

- Typically, for the MAP estimator, the larger n (the size of the data), the less important the prior $\log p_0(\theta)$ is in the overall expression

$$\log p_0(\theta) + \sum_{i=1}^n \log p(x_i | \theta)$$

→ omit the prior (?!)

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Maximum Likelihood (ML) estimator

- Omitting the prior leads to the maximum likelihood (ML) method.

$$\hat{\theta} = \operatorname{argmax}_{\theta} \sum_{i=1}^n \log(p(x_i | \theta))$$

- No usage any prior knowledge (or no knowledge) about the parameters; we seek those values that "explain" the data in the best way .
- MAP & ML are equal when the prior are equal i.e. uniform distribution

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Log likelihood

- We seek a maximum of the likelihood function, log-likelihood, or any other monotonically increasing function.
- The term $\log p(X^{(n)} | \theta)$ is the log likelihood of the parameter in regards to the sample $x_1 \dots x_n$

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ML Example I

- ML estimator for the parameter of the exponential density function $p(x | \theta) = \frac{1}{\theta} \cdot e^{-\frac{x}{\theta}}$:

$$\hat{\theta} = \arg \max_{\theta} p(X^{(n)} | \theta) = \arg \max_{\theta} \prod_{i=1}^n \frac{1}{\theta} e^{-\frac{x_i}{\theta}}$$

$$= \arg \max_{\theta} \sum_{i=1}^n \ln \left(\frac{1}{\theta} e^{-\frac{x_i}{\theta}} \right) = \arg \max_{\theta} \sum_{i=1}^n -\frac{x_i}{\theta} - \ln \theta$$

- The maximum is achieved where

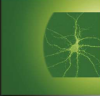
$$\frac{d \left(\sum_{i=1}^n -\frac{x_i}{\theta} - \ln \theta \right)}{d\theta} = \sum_{i=1}^n \left(\frac{x_i}{\theta^2} - \frac{1}{\theta} \right) = \frac{1}{\theta^2} \sum_{i=1}^n x_i - \frac{n}{\theta} = 0 \Rightarrow \hat{\theta} = \frac{1}{n} \sum_{i=1}^n x_i$$

→ The empirical mean (average)

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ML example II

- Let us find the ML estimator for $p(x | \theta) = \frac{1}{2} e^{-|x-\theta|}$

$$\hat{\theta} = \arg \max_{\theta} p(X^{(n)} | \theta) = \arg \max_{\theta} \prod_{i=1}^n \frac{1}{2} e^{-|x_i-\theta|} = \arg \max_{\theta} \sum_{i=1}^n \ln \left(\frac{1}{2} e^{-|x_i-\theta|} \right)$$

- Observe:

$$\ln \left(\frac{1}{2} e^{-|x_i-\theta|} \right) = -|x_i-\theta| - \ln 2 \Rightarrow \frac{d \ln \left(\frac{1}{2} e^{-|x_i-\theta|} \right)}{d\theta} = \text{sign}(x_i - \theta)$$

- The maximum is at $\hat{\theta}$ where $\sum_{i=1}^n \text{sign}(x_i - \hat{\theta}) = 0$

→ The median of the sampled data.

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MLE example III: discrete choices

Example: Assuming that upon tossing a coin 80 times we get 50 tails, which coin do we have?

- Unbiased
- Biased (2/3 tails)
- Biased (1/3 tails)

$$\ln p(D|p(T) = \frac{1}{3}) = \ln \prod_{k=0}^{80} p(x_k|p(T) = \frac{1}{3})$$

$$= \sum_{k=0}^{80} \ln p(x_k|p(T) = \frac{1}{3})$$

$$= \ln \left(\frac{1}{3}\right)^{50} + \ln \left(\frac{2}{3}\right)^{30}$$

$$= 50 \times \ln \frac{1}{3} + 30 \times \ln \frac{2}{3}$$

$$= -67.095$$

$$\ln p(D|p(T) = \frac{1}{2}) = \ln \prod_{k=0}^{80} p(x_k|p(T) = \frac{1}{2})$$

$$= \sum_{k=0}^{80} \ln p(x_k|p(T) = \frac{1}{2})$$

$$= \ln \left(\frac{1}{2}\right)^{50} + \ln \left(\frac{1}{2}\right)^{30}$$

$$= 50 \times \ln \frac{1}{2} + 30 \times \ln \frac{1}{2}$$

$$= -55.452$$

$$\ln p(D|p(T) = \frac{2}{3}) = \ln \prod_{k=0}^{80} p(x_k|p(T) = \frac{2}{3})$$

$$= \sum_{k=0}^{80} \ln p(x_k|p(T) = \frac{2}{3})$$

$$= \ln \left(\frac{2}{3}\right)^{50} + \ln \left(\frac{1}{3}\right)^{30}$$

$$= 50 \times \ln \frac{2}{3} + 30 \times \ln \frac{1}{3}$$

$$= -53.232 \star$$

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MLE example III: optimal estimator

- Can we find the optimal parameter?

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$\ln \frac{\partial p(D|p)}{\partial p} = \ln \left(\binom{n}{x} p^x (1-p)^{n-x} \right)$$

$$= \frac{\partial p(D|p)}{\partial p} \left(\ln \binom{n}{x} + \ln p^x + \ln (1-p)^{n-x} \right)$$

$$= \frac{x}{p} - \frac{n-x}{1-p}$$

- Zero derivative $\rightarrow \frac{(1-p)x}{p} = \frac{p(n-x)}{n}$
- Thus, our MLE estimator is 50/80=0.625

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ML problem for small data sample I

- Suppose we have a coin and wish to estimate the outcome (head or tail) from observing a series of coin tosses. θ = probability of tossing a head.
- After observing n coin tosses, we note that h trials are head.
- To estimate whether the next toss will be head or tail, we form an ML estimator:

$$L(\theta) = p(x_1, \dots, x_n | \theta)$$

$$= p(x_1 | \theta) \cdots p(x_n | \theta) = \theta^h \cdot (1-\theta)^{n-h}$$

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ML problem for small data sample II

$$\log L(\theta) = h \log \theta + (n-h) \log(1-\theta)$$

$$\frac{d}{d\theta} \log L(\theta) = \frac{h}{\theta} - \frac{n-h}{1-\theta} = 0$$

$$\theta_{ML} = \frac{h}{n}$$

- After one toss, if it comes up tail, our ML estimate predicts zero probability of seeing heads. If first n tosses are tails, the ML continues to predict zero probability of seeing heads.

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Bayesian estimation I

- The Bayesian estimator may use **any** loss function.

Example 1: $\lambda(\hat{\theta}, \theta) = |\hat{\theta} - \theta|$ (absolute error)

- The total Bayesian risk here:

$$R[\hat{\theta} | X^{(n)}] = \int_{-\infty}^{\infty} p(\theta | X^{(n)}) |\hat{\theta} - \theta| d\theta$$

$$= \int_{-\infty}^{\hat{\theta}} p(\theta | X^{(n)}) (\hat{\theta} - \theta) d\theta + \int_{\hat{\theta}}^{\infty} p(\theta | X^{(n)}) (\theta - \hat{\theta}) d\theta$$

- Minimum:

$$\frac{dR[\hat{\theta} | X^{(n)}]}{d\hat{\theta}} = \int_{-\infty}^{\hat{\theta}} p(\theta | X^{(n)}) d\theta - \int_{\hat{\theta}}^{\infty} p(\theta | X^{(n)}) d\theta$$

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Bayesian estimation II

- At the $\hat{\theta}$ which is a solution we have
$$\int_{-\infty}^{\hat{\theta}} p(\theta | X^{(n)}) d\theta = \int_{\hat{\theta}}^{\infty} p(\theta | X^{(n)}) d\theta$$
- That is, for the $\lambda(\hat{\theta}, \theta) = |\hat{\theta} - \theta|$ the optimal Bayesian estimator for the parameter is the median of the distribution $p(\theta | X^{(n)})$

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Bayesian estimation III

Example 2: $\lambda(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$ (squared error).

- Total Bayesian risk:

$$R[\hat{\theta} | X^{(n)}] = \int p(\theta | X^{(n)}) (\hat{\theta} - \theta)^2 d\theta$$
- Again, in order to find the minimum, let the derivative be equal 0:

$$\frac{dR[\hat{\theta} | X^{(n)}]}{d\hat{\theta}} = \int p(\theta | X^{(n)}) 2(\hat{\theta} - \theta) d\theta$$

$$= 2\hat{\theta} \int p(\theta | X^{(n)}) d\theta - 2 \int p(\theta | X^{(n)}) \theta d\theta = 2(\hat{\theta} - E[\theta | X^{(n)}]) = 0$$
- The optimal estimator here is the conditional expectation of θ given the data $X^{(n)}$.

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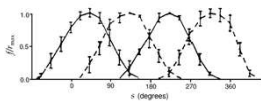
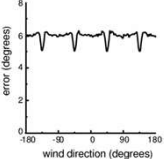
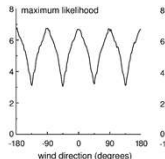
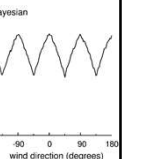


Optimal decoding

Cricket cercal system.

$$p(r|s) = \prod_{a=1}^A p_a(r_a|s)$$

with $p_a(r_a|s)$ Gaussians centered on $r_{a0}(s)$. $p(s)$ is constant.

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Outline

- Population vectors
- Optimal estimators
- Estimation error

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Errors

- Statistical error:** the difference between a computed, estimated, or measured value and the true, specified, or theoretically correct value that is caused by random, and inherently unpredictable fluctuations in the measurement apparatus.
- Systematic error:** the difference between a computed, estimated, or measured value and the true, specified, or theoretically correct value that is caused by non-random fluctuations from an unknown source, and which, once identified, can usually be eliminated.

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Estimation bias

- The estimation **bias** is the mean error of the estimator for a specific stimulus

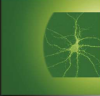
$$\text{bias}(\hat{x}) = \langle \hat{x} \rangle - x$$
- An estimator is called **unbiased** if $\text{bias}(\hat{x}) = 0$ for every stimulus
- The estimation **variance** measures how the estimator varies around its mean

$$\sigma^2(\hat{x}) = \langle (\hat{x} - \langle \hat{x} \rangle)^2 \rangle$$

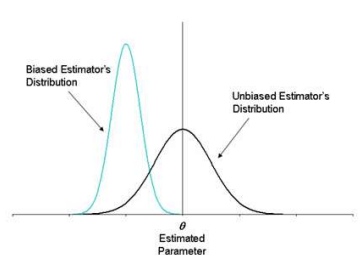
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Estimators

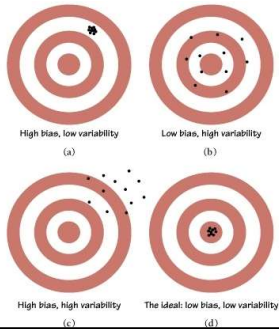


(weibull.com)

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Bias vs. Variance



(a) High bias, low variability

(b) Low bias, high variability

(c) High bias, high variability

(d) The ideal: low bias, low variability

(pinterest.com)

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Estimation error

$$Error(\hat{x}) = E[(\hat{x} - x)^2] = \sigma^2(\hat{x}) + bias^2(\hat{x})$$

- The estimation error is the sum of the estimator variance and its squared bias.

$$E((\hat{\theta} - \theta)^2) = E((\hat{\theta} - E(\hat{\theta})) + (E(\hat{\theta}) - \theta))^2 = E\{(\hat{\theta} - E(\hat{\theta}))^2 + (E(\hat{\theta}) - \theta)^2 + 2(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta)\}$$

$$= Var(\hat{\theta}) + Bias^2(\hat{\theta}) + E((\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta)) = Var(\hat{\theta}) + Bias^2(\hat{\theta}) + E(\hat{\theta}E(\hat{\theta}) - E(\hat{\theta})\theta) - \theta\hat{\theta} + E(\hat{\theta})\theta$$

$$= Var(\hat{\theta}) + Bias^2(\hat{\theta}) + (E(\hat{\theta}))^2 - (E(\hat{\theta}))^2 - \theta E(\hat{\theta}) + \theta E(\hat{\theta}) = Var(\hat{\theta}) + Bias^2(\hat{\theta}).$$

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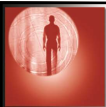
Biased estimator

- Famous example (from wikipedia.org)
- A biased estimator of the variance:
$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$
- Since:
$$E[S^2] = E\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right] = E\left[\frac{1}{n} \sum_{i=1}^n ((X_i - \mu) - (\bar{X} - \mu))^2\right]$$

$$= E\left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu) \frac{1}{n} \sum_{i=1}^n (X_i - \mu) + (\bar{X} - \mu)^2\right]$$

$$= E\left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X} - \mu)^2\right] = \sigma^2 - E[(\bar{X} - \mu)^2] < \sigma^2$$

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Unbiased estimator

- An unbiased estimator of the variance:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$
- Since:

$$E[(\bar{X} - \mu)^2] = \frac{1}{n} \sigma^2$$
- And thus:

$$\frac{n}{n-1} * [\sigma^2 - E[(\bar{X} - \mu)^2]]$$